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Navigating the landscape of Case Fidelity and Competence in Case-Based Reasoning

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Abstract. In this paper, we survey measures aimed at quantifying the intrinsic quality of cases' contents, which we collectively refer to as case *fidelity*, and measures that capture their *competence* within the Case-Based Reasoning literature. We discuss how insights from the Truth Discovery and Item Response Theory literature can respectively inform advancements in estimating case fidelity and competence. Additionally, we highlight novel research directions that emerge from a deeper examination of case fidelity and competence.

Keywords: Case-Based Reasoning, Case Base Maintenance, Case Competence

1 Introduction

Case-Based Reasoning (CBR) [7] is a problem-solving paradigm that involves solving an input problem by first retrieving cases (problem-solution pairs) similar to it from a repository called the case base. The solutions proposed by these neighboring cases are then modified using the adaptation knowledge to suit the particular needs of the problem in question. If the proposed solution proves effective for the problem at hand, this problem-solution pair is optionally retained in the case base. This reasoning ability of the case-based reasoner is attributed to the four knowledge containers, namely the vocabulary (the case representation strategy), case base, similarity, and adaptation [23].

It is known that, over time, one or more of these knowledge containers may accumulate invalid or outdated knowledge. To address this, the broader framework of Case-Based Reasoner Maintenance [31] was proposed and involves the maintenance of the four knowledge containers. In that, the maintenance of the case bases, referred to as Case Base Maintenance (CBM) [5], which typically involves the removal of *poor quality* cases from the case base, has garnered significant attention. This evaluation of case quality often encompasses various criteria, such as whether the solution present in a case is suitable for its corresponding problem, the contribution of the case in preserving the problem-solving ability of the reasoner, etc. In addition, recent developments have also highlighted the necessity to account for the contribution of cases in making unfair decisions, such as in [17].

In this work, we provide what we believe is the first of its kind, a unified view of the approaches in the CBR literature that attempt to quantify case *fidelity* — a term we use to collectively refer to measures that capture the intrinsic quality of a case’s contents — and its *predictive competence* — the criticality of a case in preserving the problem-solving ability of the reasoner. It is to be noted that our study does not center on methodologies for maintaining case bases. Readers interested in exploring recent advancements in CBM methodologies are encouraged to consult the comprehensive survey provided in [5]. Additionally, our analysis of case competence is confined to the predictive performance of cases, excluding notions such as *explanatory competence*, which pertains to explanation aspects as outlined in [6]. Henceforth, we will refer to predictive competence simply as competence.

We further explore methodologies beyond the confines of CBR to identify frameworks that may inspire novel approaches for quantifying the fidelity and competence of cases. Furthermore, our study of case fidelity and competence unveils several open research directions that include the necessity for the fidelity and competence measure to interact and the imperative need to widen the scope of maintenance to other knowledge containers as well.

The organization of our work is summarized as follows. In Section 2, we survey methodologies proposed in the literature for quantifying the fidelity and competence of cases. Section 3 explores literature beyond the realm of CBR that may offer potential insights into advancing methodologies for quantifying case fidelity and competence. Finally, in Section 4, we highlight key directions for research that spawn from a deeper inspection of case fidelity and competence.

2 Literature Review

In this section, we provide background into two classes of measures used to quantify the goodness of cases, namely, their fidelity and competence. We review the existing approaches in the literature that attempt to capture these properties. As we shall see, the demarcation between these two facets appears somewhat blurry in the literature. We finally illustrate, using a synthetic example, that although these properties may be perceived as indicators of the goodness of a case, they need not necessarily have a bearing on one another.

2.1 Understanding Fidelity and Competence

The Merriam-Webster dictionary defines the term fidelity as ‘accuracy in details’³. In the context of CBR, the *true* fidelity of a case would refer to how well a domain expert perceives its solution component appropriate for the problem description (or conversely). In this work, we refer to a case with high (low) fidelity as *faithful* (*unfaithful*).

In practice, however, obtaining expert feedback may not always be feasible or timely [9], thereby making it difficult to acquire true fidelity scores. The CBR

³ <https://www.merriam-webster.com/dictionary/fidelity>

community has proposed multiple approaches over the years to arrive at a proxy to the true fidelity estimates of cases in the case base via bottom-up means. As would be surveyed in Section 2.2, these methods largely rely on arriving at fidelity estimates of cases by inspecting the extent to which they comply with the foundational assumption in CBR that *similar problems have similar solutions*.

It is important to note that the estimates arrived at by such approaches need not necessarily conform with human assessments. Specifically, a case could be attributed a high fidelity score by an expert but not by bottom-up measures due to the presence of poor similarity and/or adaptation knowledge.

The competence of a case, on the other hand, can be defined as the *criticality* of the case in preserving the problem-solving ability of the reasoner. A case that plays an indispensable role by heavily contributing to the problem-solving ability of the reasoner is regarded as competent, while a case whose contribution is subsumed by other cases is deemed redundant.

Given that it is not possible to envisage all possible problems that the reasoner may encounter in the future, the *true* competence of cases in the case base cannot be estimated. As we shall see in Section 2.3, the measures proposed for estimating the competence of a case operate under the assumption that the case base serves as a good representative of the problems that the reasoner might encounter in the future [27]. Thus, the criticality of a case in preserving the problem-solving ability over its cases is used as a proxy for its true competence.

A distinguishing feature between fidelity and competence concerns with their implications on the effectiveness and efficiency of the reasoner. In particular, the deletion or undermining of cases with low fidelity is associated with an improvement in the prediction quality of the reasoner. Competence, on the other hand, has predominantly been used for the compaction of case base, thereby leading to efficiency enhancements. While an improvement in efficiency need not necessarily result in effectiveness gains, experiments in [13] have shown some improvements in average accuracy scores upon removal of redundant cases.

It is crucial to emphasize at this point that although fidelity and competence scores may appear to associate exclusively with the cases in the case base, they cannot be determined in isolation from other knowledge containers. For instance, the neighborhood information necessitates the similarity knowledge container, while the *solves* function also incorporates adaptation. In all the approaches that follow, the fidelity and competence estimates are considered to be implicitly conditioned on one or more of the knowledge containers.

2.2 Approaches for estimating Case Fidelity

The literature on estimating case fidelity has proposed a multitude of approaches. These approaches operate under an implicit assumption that the unfaithfulness of cases stems from their solution components. A predominant theme observed in quantifying the fidelity of cases involves inspecting the extent to which a case is *well aligned* with its neighborhood. In this section, we briefly review these measures.

Alignment-based Approaches

The *cohesion* measure [8], one of the first case fidelity measures, captures the extent of overlap between the problem-side and solution-side neighbors of a case.

$$cohesion(c) = \frac{|\mathcal{N}_c^{Prob} \cap \mathcal{N}_c^{Sol}|}{|\mathcal{N}_c^{Prob} \cup \mathcal{N}_c^{Sol}|} \quad (1)$$

where $\mathcal{N}_c^{Prob}$ and $\mathcal{N}_c^{Sol}$ represent the problem-side and solution-side neighbors of the case c , respectively. A high cohesion value is assigned to a case if its problem-side neighbors tend to have similar solutions *and* the solution-side neighbors have similar problem parts.

A significant limitation of the cohesion measure arises from the observation that, while it is necessary in CBR for problem-side neighbors to have similar solutions, the converse is not essential. Consequently, a case may be unjustly penalized by the cohesion measure if its solution-side neighbors do not exhibit similar problem-side attribute values. In contrast, the *alignment* measure proposed in [15] addresses this issue by assigning higher scores to cases whose problem-side neighbors have similar solutions while disregarding the converse. The alignment of a case c with its problem-side neighbors $\mathcal{N}_c$ is given by,

$$alignment(c) = \frac{\sum_{c' \in \mathcal{N}_c} s_{cc'}^{Prob} \times s_{cc'}^{Sol}}{\sum_{c' \in \mathcal{N}_c} s_{cc'}^{Prob}} \quad (2)$$

where $s_{cc'}^{Prob}$ and $s_{cc'}^{Sol}$ indicate respectively the problem-side and solution-side similarity⁴ between c and c' . Furthermore, the formulation allows a closer problem-side neighbor to exert more influence on the alignment score than a distant one.

It is important to note that the applicability of cohesion and alignment measures extends beyond regression and classification settings. An empirical comparison of these measures in the context of Textual CBR can be found in [22].

While it is a common practice to use the k -nearest neighbors, denoted here by $\mathcal{N}_c^k$ for the case c , Equation 2 was further extended in [2] where a higher-order neighborhood $\mathcal{N}_c^{k^n}$ was considered. For instance, $\mathcal{N}_c^{k^2}$ refers to the second-order neighbors of c and contains the k -nearest neighbors of the cases in $\mathcal{N}_c^k$.

Another approach that accounts for a wider neighborhood is the *Complexity* measure [13]. Proposed in a classification setting, it captures the proportion of neighbors of c that belong to a class different from that of c as the neighborhood size increases. The complexity within the neighborhood of c is given by,

$$complexity(c) = 1 - \frac{1}{K} \sum_{k=1}^K P_{l_c}(\mathcal{N}_c^k) \quad (3)$$

where $P_{l_c}(\mathcal{N}_c^k)$ denotes the proportion of neighbors in $\mathcal{N}_c^k$ that belong to l_c , the class label of c . A case c with complexity greater than a predefined threshold is considered unfaithful.

⁴ It is important to recognize that estimating solution-side similarity is often a non-trivial task in various settings. We point the readers to [21] for relevant discussions in the context of Textual CBR (TCBR).

The *heterogeneity* score proposed in [29] and Reputation-Based Maintenance [18], much like complexity, also account for the distribution of the neighboring cases to the different classes in the set of classes $\mathcal{L}$. The heterogeneity score associated with a case c is given by,

$$\text{heterogeneity}(c) = \sum_{l \in \mathcal{L}} P_l(\mathcal{N}_c \cup \{c\})^2 \quad (4)$$

The *reputation* of a case c is the difference between the number of neighbors with the same class as c and the number of those belonging to a different class.

$$\begin{aligned} \text{reputation}(c) &= \sum_{c' \in \mathcal{N}_c^k} \mathbb{I}(v_c = v_{c'}) - \sum_{c' \in \mathcal{N}_c^k} \mathbb{I}(v_c \neq v_{c'}) \\ &= |\mathcal{N}_c^k| \times (P_c(\mathcal{N}_c^k) - P_{\mathcal{L}-l_c}(\mathcal{N}_c^k)) \end{aligned} \quad (5)$$

where $\mathbb{I}(p)$ is the identity function that returns 1 if p evaluates to **true**, 0 otherwise. This approach regards a case as unfaithful if it has a negative reputation.

The *Friend-to-Enemy* ratio ($F : E$) [14], in contrast to the class distribution-based approaches, makes use of the average distance of the case c to its k -nearest like neighbors $\mathcal{N}_c^{k(+)}$ and that to its k -nearest unlike neighbors $\mathcal{N}_c^{k(-)}$.

$$F : E(c) = \frac{\sum_{c' \in \mathcal{N}_c^{k(+)}} d(c, c')}{\sum_{c' \in \mathcal{N}_c^{k(-)}} d(c, c')} \quad (6)$$

where $d(c, c')$ is the distance between the problem parts of c and c' . Consequently, c is regarded unfaithful if it is closer to its enemies ($\mathcal{N}_c^{k(-)}$) than its friends ($\mathcal{N}_c^{k(+)}$).

The methods discussed thus far implicitly assume that the neighboring cases of a given case provide a sound basis for assessing its fidelity. In particular, they regard all the cases within $\mathcal{N}_c$ as uniformly faithful when estimating the fidelity of c . This may result in erroneous conclusions in the presence of truly unfaithful neighbors. Specifically,

1. Consider a case c that is truly faithful but has a neighborhood predominantly comprising unfaithful cases. Since a large proportion of the cases similar to c have dissimilar solutions, c would be falsely assigned low fidelity by the measures discussed this far.
2. On the other hand, if c is a truly unfaithful case situated in a neighborhood of predominantly unfaithful cases, the problems similar to c can potentially have similar solutions. Thus, c may be falsely regarded as faithful by virtue of its agreement with its unfaithful neighbors.

That is, the fidelity of the neighboring cases in $\mathcal{N}_c$ should be accounted for while estimating the fidelity of c in order to discount the influence of unfaithful neighbors. RelCBR, proposed in [19], uses a circular definition to quantify the

reliability of cases in the case base. It considers *a case to be reliable if it is well aligned with its reliable neighbors*. The reliability vector $r^* = \{r_c^*\}_{c \in \mathcal{C}}$ is given by

$$r^* = \underset{\{r_c\}_{c \in \mathcal{C}}}{\operatorname{argmin}} \sum_{c \in \mathcal{C}} r_c \times d^{Sol}(v_c, v_c^e)^2 \quad (7)$$

where r_c is the reliability score assigned to the case c , v_c^e denotes the reliability-weighted solution proposed by cases in $\mathcal{N}_c$ for c , and $d^{Sol}(v_c, v_c^e)$ captures the discrepancy between v_c^e and the solution mentioned in c . The details associated with the optimization procedure can be found in [19].

A notable distinguishing trait of reliability, compared to other fidelity estimation approaches discussed previously, is its ability to account for the adaptation knowledge when computing the solution proposed for cases ($\{v_c^e\}_c$) using the solution of their neighbors.

We would also like to emphasize that the above circular formulation lends RelCBR visibility to a wider neighborhood. Specifically, the reliability of a case c is determined by examining the reliability of cases in $\mathcal{N}_c^k$. However, the reliability of each neighbor within $\mathcal{N}_c^k$ is itself dependent on the reliability of their own neighbors, collectively denoted by $\mathcal{N}_c^{k^2}$, and so forth. Therefore, r_c is effectively dependent on the reliability of cases in its higher-order neighborhood.

Finally, we present a comparative analysis of the fidelity estimation approaches in Table 1.

	Cohesion	Alignment	Higher-order Alignment	Complexity	Heterogeneity	Reputation	F:E	Reliability
Need for solution-side similarity/neighbors?	Y	Y	Y	N	N	N	N	Y
Local (L) or Extended (E) Visibility?	L	L	E	E	L	L	E	E
Applicable beyond Classification settings?	Y	Y	Y	N	N	N	N	Y
Accounts for adaptation knowledge?	N	N	N	N	N	N	N	Y
Agnostic to the presence of unfaithful neighbors?	Y	Y	Y	Y	Y	Y	Y	N

Table 1: A comparison of fidelity estimation approaches.

Provenance-based Approach

The alignment-based approaches have been largely employed to estimate the fidelity of cases within a static snapshot of the case base. However, the provenance-based approach introduced in [9] offers a temporal perspective, aiming to identify potentially unfaithful cases by considering the growth of the case base over time. This approach operates under the assumption that the case base comprises faithful cases during the onset of the reasoner, with the accumulation of unfaithful cases occurring due to the retention of unverified cases. The claim is that a case whose parent case is itself unverified is more likely to be unfaithful compared to another with a verified parent case. In general, the unfaithfulness of a case is captured by the length of the path to its nearest verified ancestor.

2.3 Approaches for estimating Case Competence

The competence of a case denotes its *criticality* in preserving the problem-solving ability of the reasoner. In other words, a case with low competence may be

deemed *redundant* to the reasoner's problem-solving ability. An accurate estimate of the competence of the cases in the case base would require assessing the cases over all possible scenarios that the reasoner may encounter. However, since this information is not known apriori, it is commonplace to assume that the case base denotes a representative sample of the future target problems [27]. Consequently, the competence of a case c is measured in terms of its criticality in solving the other cases from the case base.

The most rudimentary measure of competence of a case c is given by the cardinality of its *Coverage Set* [28], the cases which are *solved* by c .

$$CoverageSet(c) = \{ c' \mid c' \in \mathcal{C}, c \text{ solves } c' \} \quad (8)$$

where $c \text{ solves } c' = \mathbf{true}$ indicates that c is retrieved for c' and the solution of c can be adapted to arrive at the solution of c' .

The *Competence Measure (CM)* proposed in [1] regards a case to be competent if it solves multiple cases but is solved by few cases, and is given by

$$CompetenceMeasure(c) = \frac{|CoverageSet(c)|}{|ReachabilitySet(c)|} \quad (9)$$

where $ReachabilitySet(c)$ denotes the set of cases in the case base that solve c .

These measures regard the number of cases solved by a case as a key indicator of its competence. However, it is recognized that a case with large coverage need not necessarily be competent by virtue of the possibility of its coverage being subsumed by other cases. In contrast, a case may be regarded as competent despite having a small coverage set comprising of cases that are not solved by other cases in the case base.

The *Relative Coverage* [27] attempts to address this issue by rewarding a case for solving cases that are not solved by many cases.

$$RelativeCoverage(c) = \sum_{c' \in CoverageSet(c)} \frac{1}{|ReachabilitySet(c')|} \quad (10)$$

This measure of case competence is also used to identify the *representative* cases in each competence group using the footprint algorithm [27]. Other approaches to identify representative cases make use of clustering techniques, such as in [26].

The Relative Coverage provides yet another perspective in the context of CBM. Heuristically, it denotes the likelihood of a case being retained in the case base post CBM. This interpretation of case competence is explored in the *Retention Score* measure [16]. It further addresses a key limitation of the Relative Coverage measure by allowing Compositional Adaptation [30].

$$RS(c) = \sum_{c' \in CoveredCases(c)} RS(c') \sum_{\mathbb{C} \in SupportCases(c, c')} \min_{c' \in \mathbb{C}} \frac{RS(c')}{|\mathbb{C}|} \quad (11)$$

In essence, a case c is assigned a high retention score if (i). c participates in solving multiple cases (captured by *Covered Cases*) that have high Retention Score, (ii). c requires support from a few cases to solve $c' \in CoveredCases(c)$ (captured by each $\mathbb{C} \in SupportCases(c, c')$), and (iii). These supporting cases are likely to be retained.

2.4 Relation between Fidelity and Competence

Since fidelity and competence capture the goodness of the cases in the case base, a natural question in this context is whether these characteristics have a bearing on one another. Specifically, if information regarding a case's fidelity is available, can we infer anything about its competence, or vice versa?

Consider a case-based reasoner that predicts the class of a test instance as that of its nearest neighbor (i.e., null adaptation) based on the Euclidean distance measure from its case base depicted in Figure 1. In such a setting, it may be seen that the cases within the case base may be categorized as both faithful and competent (such as c_1 and c_2 that can collectively preserve the problem-solving ability of the reasoner to a large extent), unfaithful and redundant (c_3), and faithful but redundant (c_6). However, there is also a possibility of a case being perceived as unfaithful but competent. Cases c_4 and c_5 may be considered unfaithful but can be regarded as competent by virtue of them exclusively solving each other, thereby preserving the problem-solving ability of the reasoner over its cases. It is important to note that the notion of competence does not inherently account for the fidelity of cases, and as a result, c_4 and c_5 , although likely unfaithful, are attributed high competence.

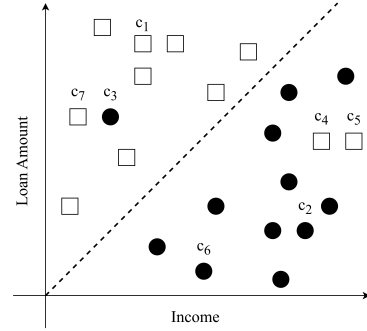


Fig. 1: A sample case base in the Loan Application domain. Each case has two attributes *Income* and *Loan Amount* in its problem part, while the solution is either *Approved* (denoted by ●) or *Rejected* (denoted by □).

This example illustrates that it is possible for a case to have high or low fidelity irrespective of its competence, and vice versa. This underscores the inadequacy of any atomic measure - one that cannot be broken down into two or more parts - to simultaneously capture both the fidelity and competence of cases within the case base. This, in our opinion, demands a reassessment in the literature, where approaches such as complexity (Equation 3) and reputation (Equation 5) claim to capture both aspects simultaneously.

3 Literature beyond CBR for estimating Fidelity and Competence

In this section, we highlight relevant literature outside the CBR context that may inspire further development in estimating the fidelity and competence of cases in the case base. We also highlight why these may not be trivially applicable in the CBR context by bringing out how they fundamentally differ from CBR. The following discussion also motivates the necessity to propose novel circularity-aware approaches [20] for estimating case fidelity and competence.

3.1 Truth Discovery

Truth Discovery [11] deals with a set of *sources* that provide potentially conflicting *answers* to *questions* from a set of questions. Within this framework, the objective is to estimate the *reliability* of the sources and *trustworthy* answers to the questions. In contrast to assigning uniform reliability to the sources and considering the majority voted answer as trustworthy, Truth Discovery advocates for a nuanced reliability-weighted aggregation.

The overarching theme for estimating the two sets of parameters - reliability of sources and the trustworthy answers to the questions - is captured in the following circular statement:

A source is considered reliable if it provides trustworthy answers. An answer is deemed trustworthy if it is supported by multiple reliable sources.

Although mapping this setting to the CBR context might appear trivial, there are complexities involved. We highlight the key differences here. For an input query (question) q , the similarity function dictates an eligibility criterion that permits only a few cases to participate in answering q . Further, the solution proposed by each of the eligible cases c is not directly projected as a proposed solution for q but is instead passed through the adaptation function. In addition, cases may be allowed to collaborate by means of compositional adaptation, unlike the typical independent source assumption in Truth Discovery. In effect, the reliability score thus obtained is a commentary not only on a case in isolation but also accounts for the similarity function and adaptation rule employed.

One approach, motivated by Truth Discovery, is proposed in [19] with a focus on regression and classification tasks. Further extensions could explore settings beyond classification and regression, examine the algorithm’s convergence guarantees, and incorporate aspects such as fine-grained source reliability [11].

3.2 Item Response Theory

The Item Response Theory (IRT) [4] is widely employed in the context of Psychometrics for assessing the latent *abilities* of the respondents and the *difficulty* of the question⁵ presented to the respondents. The use of IRT in the evaluation of high-stakes examinations such as the Scholastic Assessment Test (SAT) and Graduate Record Examinations (GRE) underscores its substantive influence. IRT has also recently garnered attention in the context of Natural Language Processing [24] and more broadly in Machine Learning [12].

IRT allows discrimination between two participants even though they correctly answer the exact same number of questions in a questionnaire when the correct responses by one are attributed to a set of simpler questions while the other correctly answers more difficult ones.

⁵ Technically, referred to as *items* in IRT.

Note the departure from the Truth Discovery setting in that each question is marked with a ground truth solution, but the difficulty annotations are unavailable. The two properties - respondent ability and item difficulty - are intertwined:

A respondent who answers multiple difficult questions, in addition to the simpler ones, will be assigned a higher competence, while questions answered primarily by competent respondents are regarded as difficult.

In the context of CBR, competence estimation approaches based on *CoverageSet* (Equation 8 and 9) account solely for the number of cases (questions) solved. Relative Coverage (Equation 10), on the other hand, considers a slightly nuanced perspective. It models an aspect similar to the difficulty of q , the case being solved, by means of the $1/|ReachabilitySet(q)|$ term. That is, a case is deemed difficult if it is solved by fewer (competent or otherwise) cases.

It is important to note that the Relative Coverage measure persists to have setbacks despite its significant resemblance to the IRT framework. Firstly, the difficulty measure used in Relative Coverage remains competence agnostic, unlike the circular statement employed in IRT. Further, its applicability is restricted to reasoners that employ 1-nearest neighbor.

4 Potential Research Directions

In this section, we enumerate some research directions that emerge from our discussions thus far. In particular, the significant deficit of the consideration of adaptation knowledge while computing case fidelity can be traced down to how the problem-solution regularity [10] is defined in the literature. We propose the need for a reassessment of this in the light of adaptation knowledge. Secondly, insights from Section 2.4 motivate the need for interaction between fidelity and competence. Lastly, we advocate for extending the focus of maintenance efforts to include not only the case base but also other knowledge containers.

4.1 An Adaptation-aware definition of Problem-Solution Regularity

Validating the applicability of CBR in a given problem domain involves verifying whether the foundational assumption that similar problems yield similar solutions largely holds [13]. Measures such as cohesion (Equation 1) and alignment (Equation 2) have been used in the literature to quantify this problem-solution regularity within the local neighborhood of a case. An average of these across all the cases in the case base gives an estimate of the problem-solution regularity across the entire case base. If this global regularity measure falls below a predefined threshold, it often suggests that CBR may not be suitable for the task. Additionally, the regularity measures can also support authoring case-based reasoners [8] by assisting in making a choice between different configurations of knowledge containers.

While measures like cohesion and alignment have been employed in designing case-based reasoners, their limitations have been acknowledged in the literature.

As noted by the authors in [3], measures such as cohesion and alignment fail to account for adaptation. However, experimental results in [3] indicate that incorporating adaptation into the computation of global problem-solution regularity results in better correlation with generalization accuracy compared to adaptation-agnostic measures.

This motivates the need for a shift in perspective from viewing problem-solution regularity as merely similar problems having similar solutions. Instead, we suggest an adaptation-aware viewpoint by accounting for all four knowledge containers.

A case c is well-aligned with its neighbors when the solutions of the neighbors can be adapted to solve c .

This definition of problem-solution regularity offers two key benefits: firstly, it can allow for better estimates of the suitability of CBR for a given problem setting and can assist in authoring CBR systems. Secondly, it can serve as a sound basis for further advancements in case fidelity estimation techniques that encompass adaptation knowledge.

4.2 Interaction between Fidelity and Competence computation

In the CBR literature, fidelity and competence computation are typically viewed as non-interacting steps. However, the case of c_4 and c_5 from Figure 1 being deemed competent while potentially being unfaithful motivates the need for an interaction between the two. Potential strategies for such an interaction involve (i). performing fidelity assessment followed by a fidelity-aware competence computation, (ii). computing competence scores and incorporating them into the fidelity assessment, or (iii). iteratively cycling through fidelity and competence computations.

The experiments in [13] propose discarding unfaithful cases before computing the competence of cases (akin to strategy (i) noted above) to prevent assigning high competence scores to cases that have noisy class labels. While this suggests that fidelity assessment is a useful precursor for competence computation, the exploration of other configurations remains open in the literature.

We note that the deletion of unfaithful cases prior to estimating competence in [13] disregards the fact that cases exhibit varying degrees of fidelity rather than being strictly faithful or unfaithful. This may lead to the loss of potentially useful information. An alternate strategy might be to incorporate fidelity awareness in the estimation of the competence of cases. A similar extension - competence-aware fidelity - may be proposed for the other potential interaction strategy.

4.3 Extending Maintenance to other Knowledge Containers

One central bottleneck in effective CBR maintenance is the availability of expert knowledge. In the spirit of active learning, it makes sense to selectively seek expert intervention in a way that can potentially have the highest impact on the

problem-solving ability of the reasoner. This suggests that the order in which we present cases or other knowledge containers to an expert to seek their feedback can play an important role in minimizing the overall knowledge acquisition effort.

While CBM has received much attention in the context of the maintenance of case-based reasoners, the maintenance of knowledge containers may warrant closer inspection as well. For instance, a lack of problem-solution regularity between cases in a certain locality can be attributed to deficiencies in the way vocabulary, similarity (or adaptation⁶) knowledge have been encoded – and fixing one of these problems can avoid examination of several cases in isolation.

From a cognitive standpoint, the issue of CBM opens up many interesting frontiers of exploration. For a start, a case that has a solution component very different from other cases in its neighborhood appears to be an outlier and may be considered as a candidate for deletion. However, as humans, any experience that is strikingly different from the norm often paves the way for learning. Of particular mention is the classical model of human reminding proposed by Roger Schank in his seminal work *Dynamic Memory* [25] that was instrumental in paving the way for a large class of computational models of human memory and their applied counterparts, such as Case-Based Reasoning. According to Schank’s model, humans generate expectations based on generalized representations of memory, and when these expectations fail, they are curious and seek explanations for such failures. These explanations, in turn, lead to further generalizations. Thus, according to Schank’s model, a case that is not well-aligned with its neighborhood could be potentially very interesting since it could trigger expert intervention in meaningful ways to bring about changes in either the case representation itself or in associated vocabulary, similarity, or adaptation knowledge. One of Schank’s examples is the case where one visits a restaurant where one has to pay before they eat – this is strikingly different from his past experiences where people pay only after eating. Accommodating such *outliers* may necessitate revisiting the underlying vocabulary, the language in which cases are expressed and indexed. For instance, adding more attributes can help better discriminate between classes – interestingly, this is also the philosophy behind bottom-up approaches such as Support Vector Machines. In the context of CBR, attributes may be drawn from more involved types, such as trees/hierarchies or graphs, to bring about richer representations of the underlying domain knowledge.

5 Conclusion

In this work, we have presented a survey of approaches in the CBR literature aimed at quantifying the fidelity and competence of cases. We highlight how insights from Truth Discovery and Item Response Theory can serve as potential inspirations for proposing circularity-aware approaches that attempt to estimate the fidelity and competence of cases. We finally identify three key research directions emerging from a closer examination of these measures. First, we advocate

⁶ In the extended notion of alignment presented in this paper where adaptation knowledge is integrated

for a reassessment of how problem-solution regularity is defined in the literature. Second, we emphasize the need for an interaction between fidelity and competence measures. Finally, we underscore the need for further study on the maintenance of the other knowledge containers for effective CBR maintenance.

References

1. Chebel-Morello, B., Haouchine, M.K., Zerhouni, N.: Case-based maintenance: Structuring and incrementing the case base. *Knowledge-Based Systems* **88**, 165–183 (2015)
2. Dileep, K., Chakraborti, S.: Towards higher order complexity measures for text classification
3. Ganesan, D., Chakraborti, S.: A reachability-based complexity measure for case-based reasoners. In: *The Thirty-Second International Flairs Conference* (2019)
4. Hambleton, R.K., Swaminathan, H.: *Item response theory: Principles and applications*. Springer Science & Business Media (2013)
5. Juarez, J.M., Craw, S., Lopez-Delgado, J.R., Campos, M.: Maintenance of case bases: current algorithms after fifty years. *IJCAI International Joint Conferences on Artificial Intelligence Organization* (2018)
6. Keane, M.T., Smyth, B.: Good counterfactuals and where to find them: A case-based technique for generating counterfactuals for explainable ai (xai). In: *Case-Based Reasoning Research and Development: 28th International Conference, ICCBR 2020, Salamanca, Spain, June 8–12, 2020, Proceedings* 28. pp. 163–178. Springer (2020)
7. Kolodner, J.L.: An introduction to case-based reasoning. *Artificial intelligence review* **6**(1), 3–34 (1992)
8. Lamontagne, L.: Textual cbr authoring using case cohesion. In: *Proceedings of 3rd Textual Case-Based Reasoning Workshop at the 8th European Conf. on CBR*. pp. 33–43 (2006)
9. Leake, D., Whitehead, M.: Case provenance: The value of remembering case sources. In: *International Conference on Case-Based Reasoning*. pp. 194–208. Springer (2007)
10. Leake, D.B., Wilson, D.C.: When experience is wrong: Examining cbr for changing tasks and environments. In: *International Conference on Case-Based Reasoning*. pp. 218–232. Springer (1999)
11. Li, Y., Gao, J., Meng, C., Li, Q., Su, L., Zhao, B., Fan, W., Han, J.: A survey on truth discovery. *ACM Sigkdd Explorations Newsletter* **17**(2), 1–16 (2016)
12. Martínez-Plumed, F., Prudêncio, R.B., Martínez-Usó, A., Hernández-Orallo, J.: Making sense of item response theory in machine learning. In: *ECAI 2016*, pp. 1140–1148. IOS Press (2016)
13. Massie, S., Craw, S., Wiratunga, N.: Complexity profiling for informed case-base editing. In: *Advances in Case-Based Reasoning: 8th European Conference, ECCBR 2006 Fethiye, Turkey, September 4-7, 2006 Proceedings* 8. pp. 325–339. Springer (2006)
14. Massie, S., Craw, S., Wiratunga, N.: When similar problems don't have similar solutions. In: *Case-Based Reasoning Research and Development: 7th International Conference on Case-Based Reasoning, ICCBR 2007 Belfast, Northern Ireland, UK, August 13-16, 2007 Proceedings* 7. pp. 92–106. Springer (2007)

15. Massie, S., Wiratunga, N., Craw, S., Donati, A., Vicari, E.: From anomaly reports to cases. In: International Conference on Case-Based Reasoning. pp. 359–373. Springer (2007)
16. Mathew, D., Chakraborti, S.: A generalized case competence model for case base maintenance. *AI Communications* **30**(3-4), 295–309 (2017)
17. Mitra, S., Mathew, D., P, D., Chakraborti, S.: Group fairness in case-based reasoning. In: International Conference on Case-Based Reasoning. pp. 217–232. Springer (2023)
18. Nakhjiri, N., Salamó, M., Sànchez-Marrè, M.: Reputation-based maintenance in case-based reasoning. *Knowledge-Based Systems* **193**, 105283 (2020)
19. Parsodkar, A.P., P, D., Chakraborti, S.: Never judge a case by its (unreliable) neighbors: Estimating case reliability for cbr. In: International Conference on Case-Based Reasoning. pp. 256–270. Springer (2022)
20. Parsodkar, A.P., P, D., Chakraborti, S.: The case for circularities in case-based reasoning. In: International Conference on Case-Based Reasoning. pp. 85–101. Springer (2023)
21. Raghunandan, M., Chakraborti, S., Khemani, D.: Robust measures of complexity in tcbr. In: Case-Based Reasoning Research and Development: 8th International Conference on Case-Based Reasoning, ICCBR 2009 Seattle, WA, USA, July 20-23, 2009 Proceedings 8. pp. 270–284. Springer (2009)
22. Raghunandan, M., Wiratunga, N., Chakraborti, S., Massie, S., Khemani, D.: Evaluation measures for tcbr systems. In: Advances in Case-Based Reasoning: 9th European Conference, ECCBR 2008, Trier, Germany, September 1-4, 2008. Proceedings 9. pp. 444–458. Springer (2008)
23. Richter, M.M.: The Knowledge Contained in Similarity Measures. Invited Talk at the First International Conference on Case-Based Reasoning, ICCBR’95, Sesimbra, Portugal (1995)
24. Rodriguez, P., Barrow, J., Hoyle, A.M., Lalor, J.P., Jia, R., Boyd-Graber, J.: Evaluation examples are not equally informative: How should that change nlp leaderboards? In: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers). pp. 4486–4503 (2021)
25. Schank, R.C.: Dynamic Memory: A Theory of Reminding and Learning in Computers and People. Cambridge University Press, USA (1983)
26. Smiti, A., Elouedi, Z.: Wcoid-dg: An approach for case base maintenance based on weighting, clustering, outliers, internal detection and dbsan-gmeans. *Journal of computer and system sciences* **80**(1), 27–38 (2014)
27. Smyt, B., McKenna, E.: Footprint-based retrieval. In: International Conference on Case-Based Reasoning. pp. 343–357. Springer (1999)
28. Smyth, B., McKenna, E.: Modelling the competence of case-bases. In: European Workshop on Advances in Case-Based Reasoning. pp. 208–220. Springer (1998)
29. Wang, K., Liu, J.N., Ma, W.m.: A study on the reliability of case-based reasoning systems. In: 2008 IEEE International Conference on Data Mining Workshops. pp. 60–68. IEEE (2008)
30. Wilke, W., Bergmann, R.: Techniques and knowledge used for adaptation during case-based problem solving. In: Tasks and Methods in Applied Artificial Intelligence: 11th International Conference on Industrial and Engineering Applications of Artificial Intelligence and Expert Systems IEA-98-AIE Benicàssim, Castellón, Spain, June 1–4, 1998 Proceedings, Volume II 11. pp. 497–506. Springer (1998)
31. Wilson, D.C., Leake, D.B.: Maintaining case-based reasoners: Dimensions and directions. *Computational Intelligence* **17**(2), 196–213 (2001)